Uchaaran

A little practice. A deeper connection.

Uchaaran Helper · Kids Mode for Sundar Gutka

Team 12 · Khalis Foundation

Avi Oberoi · Gurjaap Kalra
Mohkm Singh · Gurasees Bajaj

The problem we are solving

Children need help pronouncing Gurbani and a reason to return to daily practice. Conversations with Khalis focused our design on two needs: engagement that sustains practice, and understanding that goes beyond repeating words.

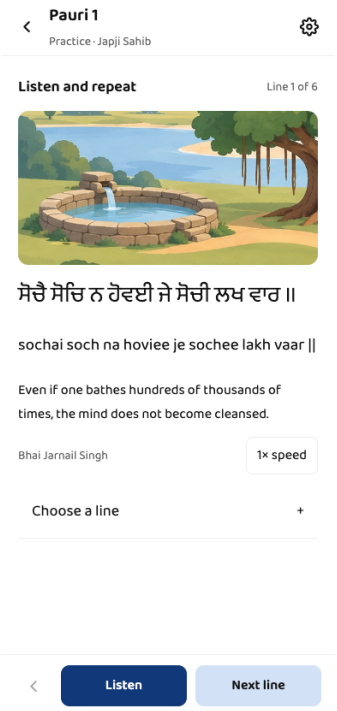
Two modes. One path toward mastery.

1. Practice at your pace

Listen and repeat with Gurmukhi, Roman-letter pronunciation and English meaning. Use recall mode to practice from memory.

2. Take the pauri challenge

Recite a line and tap Done. Azure returns word-level feedback. Revisit difficult words across 38 pauris; stars and Nanak coins unlock character rewards.



What changed since midterm

Organizers asked for less pressure and simpler feedback. We added line-by-line scoring, short word prompts, and an ML maintenance guide for Khalis.

Rooted in the Panth

Built for children, initially grades 6 to 8. Gurmukhi and the ustaad's reference recitation stay central. Familiar landmarks form a memory palace for recall; Sikh-inspired characters and seva stories connect learning with community.

Tools and implementation

Expo · React Native · TypeScript · Azure-hosted IndicConformer scoring · BaniDB text · reference audio. Codex and Paper supported development and design.

Private processing, by design

Recordings are processed privately in Azure. Temporary server recordings are deleted after scoring, with no use for public-model training.

Khalis can host the model in its own shared AWS or Azure environment. On-device packaging is another option for fully local processing that keeps recordings on device.

Ready for handover

MIT source, setup and ML maintenance docs are included. Khalis supplies hosting and credentials; costs depend on usage. An ongoing technical owner remains to be named.

Watch demo ↗ | Try the app ↗ | View code ↗ | Design process ↗